

GAYLORD, MI, USA

WMO: 725407

Lat: 45.013N

Lon: 84.701W

Elev: 1335

StdP: 14.00

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04854

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-0.6	-11.4	3.1	-4.5	-6.5	4.1	0.9	24.7	21.2	22.5	21.8	2.5	60	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	86.1	69.2	83.0	68.0	80.7	66.6	72.6	81.3	70.7	78.9	69.0	76.6	11.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.0	116.0	77.1	68.0	108.2	74.7	66.2	101.4	73.1	37.1	80.9	35.3	78.4	33.9	76.8	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	19.0	17.0	DB	-17.6	91.5	6.6	2.4	-22.3	93.2	-26.2	94.5	-29.8	95.9	-34.6	97.6
			WB	-17.9	75.3	6.5	1.9	-22.6	76.6	-26.4	77.7	-30.0	78.7	-34.7	80.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	18.4	18.8	28.0	41.1	53.8	62.8	67.4	65.5	58.8	46.3	35.0	24.2	(6)
	DBStd	19.79	10.72	10.63	11.91	10.03	9.16	7.34	6.20	5.98	8.19	9.03	9.45	9.50	(7)
	HDD50	4367	981	873	688	301	62	3	0	0	16	185	458	800	(8)
	HDD65	8178	1446	1293	1146	718	366	126	47	67	220	583	901	1265	(9)
	CDD50	1985	0	0	7	34	181	387	539	479	279	72	7	0	(10)
	CDD65	322	0	0	1	1	20	61	120	82	33	4	0	0	(11)
	CDH74	2811	0	0	6	21	211	569	1073	648	256	27	0	0	(12)
	CDH80	673	0	0	0	2	41	140	277	157	53	3	0	0	(13)
(14) Wind	WSAvg	7.4	8.1	7.8	7.2	7.7	7.5	6.7	6.4	6.1	6.7	7.5	8.2	8.3	(14)
(15) Precipitation	PrecAvg	31.9	2.2	1.5	1.6	2.4	3.1	3.2	2.9	3.5	3.4	3.2	2.6	2.2	(15)
	PrecMax	38.6	4.3	2.6	5.0	4.1	4.9	5.9	5.9	7.0	6.3	7.2	4.8	4.3	(16)
	PrecMin	26.2	0.6	0.4	0.1	0.7	0.9	0.5	1.2	1.3	0.9	0.9	0.9	0.4	(17)
	PrecStd	3.0	1.0	0.6	1.1	0.9	0.9	1.4	1.2	1.5	1.2	1.5	1.1	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	49.7	69.9	77.0	84.2	88.2	90.1	88.6	86.0	77.7	64.9	52.4	(19)
		MCWB	41.8	44.8	57.3	58.6	66.3	70.8	71.9	70.4	68.1	64.2	53.5	50.0	(20)
	2%	DB	40.3	42.9	56.0	70.0	80.0	83.9	86.1	84.2	80.8	71.5	57.4	43.2	(21)
		MCWB	37.4	38.2	47.6	54.9	63.3	67.9	69.7	68.8	66.4	61.8	50.6	40.7	(22)
	5%	DB	36.2	37.2	50.1	63.9	75.2	80.7	82.8	80.8	76.3	65.8	53.4	39.2	(23)
		MCWB	34.0	34.4	43.7	50.9	60.0	66.4	68.6	67.2	64.4	58.4	48.3	37.0	(24)
	10%	DB	33.6	33.4	44.5	58.1	70.8	76.8	80.3	77.3	72.8	60.9	48.4	36.3	(25)
		MCWB	32.1	31.1	39.1	47.2	58.4	63.4	67.0	64.8	63.1	54.7	44.1	34.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	45.2	58.4	62.0	69.6	73.8	75.3	74.3	72.0	68.1	56.4	51.4	(27)
		MCDB	44.5	47.5	67.8	70.3	80.1	84.1	85.5	82.6	79.7	74.7	61.9	52.2	(28)
	2%	WB	37.5	38.6	50.5	57.2	66.4	70.6	73.2	71.3	69.3	63.6	52.5	41.9	(29)
		MCDB	40.3	41.9	54.7	67.5	75.2	80.5	81.8	79.6	75.9	68.6	56.8	42.9	(30)
	5%	WB	34.2	34.6	44.1	53.0	63.6	68.4	71.2	69.4	67.3	58.9	48.3	37.5	(31)
		MCDB	35.9	37.3	48.2	61.8	71.7	76.4	78.8	77.2	72.9	65.4	52.5	39.0	(32)
	10%	WB	31.8	31.1	39.2	48.2	60.3	66.4	69.2	67.7	64.9	55.1	44.4	34.4	(33)
		MCDB	33.3	33.1	44.2	56.8	68.1	73.8	76.8	74.1	71.1	60.0	48.0	35.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	15.2	17.9	19.3	21.0	20.4	20.5	19.7	19.7	16.0	12.8	10.6	(35)
		MCDBR	13.3	17.7	22.4	28.7	26.7	24.1	23.4	23.5	23.4	23.4	20.2	13.5	(36)
	5% WB	MCWBR	11.4	14.5	15.0	17.3	13.5	11.7	10.8	11.1	11.8	14.0	14.4	12.0	(37)
		MCWBR	12.7	16.1	20.4	25.3	21.7	19.7	19.1	19.0	18.7	20.6	18.9	12.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.281	0.286	0.306	0.339	0.365	0.383	0.381	0.372	0.359	0.326	0.294	0.268	(40)
		taud	2.308	2.286	2.406	2.378	2.363	2.331	2.355	2.375	2.411	2.503	2.524	2.426	(41)
	Ebn at Noon		263	285	295	292	286	280	280	278	273	269	262	260	(42)
		Edh at Noon	25	31	32	36	38	39	38	36	32	25	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		387	680	1129	1452	1757	1920	1937	1646	1279	756	416	291	(44)
	RadStd		39	73	125	142	156	132	83	99	114	89	59	29	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page